

Question 6 beyond Hilbert spaces: a Hilbert-separating factor criterion

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Status. Substantial partial result, likely valid, pending human review. The arbitrary nonseparable case remains open.

1 Source question

Thieu and Yen [1] ask the following. Let E be a real normed space and let $F \subset E$ be an infinite-dimensional linear subspace. Do there exist a continuous map $\psi : E \rightarrow E^*$ and a bounded linear map $\varphi : E \rightarrow E^*$ such that

$$\text{for every } y \in E \text{ some } x \in F \text{ has } \psi(y)(x) \neq 0, \quad (\text{a3})$$

$$\text{for every } x \in F \setminus \{0\} \text{ some } y \in E \text{ has } \varphi(y)(x) \neq 0, \quad (\text{b3})$$

and

$$\sup_{y \in E} \|\varphi(y) - \psi(y)\|_{E^*} < \infty? \quad (1)$$

They prove this when E is Hilbert and state that the non-Hilbert case remains open.

We use their following theorem [1, Theorem 4.5]: every infinite-dimensional normed space M admits a continuous map $h : M \rightarrow M$ such that

$$h(m) \neq 0 \quad (m \in M), \quad \sup_{m \in M} \|h(m) - m\| \leq 1. \quad (2)$$

2 Proof intuition

The Hilbert structure is needed only to manufacture a positive bilinear form, not on the ambient space itself. If a bounded operator sends E into a Hilbert space and is injective on F , then $\varphi = T^*T$ has two key properties. It separates every nonzero vector of F , and no nonzero element of its range can vanish identically on F . The range $M = \varphi(E)$ is therefore a normed space whose nonzero vectors are already good for condition (a3). Applying (2) inside M and composing with φ removes the sole zero without changing φ by more than one.

3 The Hilbert-factor criterion

Theorem 1. *Let E be a real normed space and $F \subset E$ an infinite-dimensional linear subspace. Suppose there are a real Hilbert space H and a bounded linear operator $T_0 : E \rightarrow H$ whose restriction to F is injective. Then Question 6 has an affirmative answer. In fact, φ and ψ can be chosen so that the supremum in (1) is at most 1.*

From this and (26) we can infer that the equality (25) is valid.

The proof is complete. $\square$

4.5 Question 6

It is of interest to know what happen if the condition $\dim(F) < \infty$ in Question 5 and in Theorem 4.4 is violated. Probably, this is one of the strong motivations for the following question of Professor Ricceri [8].

Question 6: *Let E be a normed space and $F \subset E$ a linear subspace with $\dim(F) = \infty$. Does there exist a continuous mapping $\psi : E \rightarrow E^*$ and a linear and continuous operator $\varphi : E \rightarrow E^*$ satisfying the conditions*

- (a3) *For every $y \in E$ there exists $x \in F$ satisfying $\psi(y)(x) \neq 0$;*
- (b3) *For every $x \in F \setminus \{0\}$ there exists $y \in E$ satisfying $\varphi(y)(x) \neq 0$*

such that the inequality

$$\sup_{y \in E} \|\varphi(y) - \psi(y)\|_{E^*} < +\infty \quad (27)$$

holds true?

To solve the above question in the affirmative in one special case, we need the next auxiliary result on nonvanishing continuous operators having finite maximum displacements. Note that the continuous function ψ in Question 6 maps E to the dual space E^* , while the function ψ in the forthcoming theorem maps E to E .

Theorem 4.5 *Let E be an infinite-dimensional normed space. Then, there exists a nonvanishing continuous operator $\psi : E \rightarrow E$ such that*

Figure 1: Question 6 in arXiv:2502.00746v1, page 20.

Proof. Let $G = \overline{T_0(F)} \subset H$, let $P_G : H \rightarrow G$ be the orthogonal projection, and put $T = P_G T_0 : E \rightarrow G$. For $x \in F$ one has $T_0 x \in G$, hence $Tx = T_0 x$. Thus $T|_F$ is injective and $T(F)$ is dense in G .

Define

$$\varphi = T^* T : E \longrightarrow E^*.$$

For $x, y \in E$,

$$\varphi(y)(x) = \langle Ty, Tx \rangle_G. \quad (3)$$

If $x \in F \setminus \{0\}$, then

$$\varphi(x)(x) = \|Tx\|_G^2 > 0.$$

This proves (b3), with $y = x$. It also shows that $\varphi|_F$ is injective, so the normed subspace

$$M := \varphi(E) \subset E^*$$

is infinite-dimensional.

We next record the decisive range property:

$$m \in M \setminus \{0\} \implies m|_F \neq 0. \quad (4)$$

Indeed, write $m = \varphi(y)$. If $m(x) = 0$ for every $x \in F$, then (3) says that Ty is orthogonal to $T(F)$. Since $Ty \in G$ and $T(F)$ is dense in G , this forces $Ty = 0$, whence $m = T^* Ty = 0$. This proves (4).

Apply the source theorem (2) to the infinite-dimensional normed space M (no closedness of M is required) and obtain a continuous nonvanishing $h : M \rightarrow M$ with $\|h(m) - m\| \leq 1$. Set

$$\psi = h \circ \varphi : E \longrightarrow M \subset E^*.$$

For every $y \in E$, $\psi(y)$ is a nonzero member of M , so (4) supplies an $x \in F$ with $\psi(y)(x) \neq 0$. Thus (a3) holds. Finally,

$$\|\psi(y) - \varphi(y)\|_{E^*} = \|h(\varphi(y)) - \varphi(y)\|_M \leq 1,$$

which proves (1) with the asserted bound. $\square$

4 Countably total and separable subspaces

A family $(f_n) \subset F^*$ is *total* if $f_n(x) = 0$ for every n implies $x = 0$.

Corollary 2. *Question 6 has an affirmative answer whenever F^* contains a countable total family. In particular, it has an affirmative answer whenever F is separable. The subspace F need not be closed or complemented, and E need not be complete or Hilbert.*

Proof. Let $(f_n)_{n \geq 1} \subset F^*$ be total. By Hahn–Banach, extend each f_n to $\tilde{f}_n \in E^*$ without changing its norm. Put

$$a_n = \frac{2^{-n}}{1 + \|\tilde{f}_n\|}, \quad T_0 y = (a_n \tilde{f}_n(y))_{n \geq 1} \in \ell_2.$$

Then

$$\|T_0 y\|_2^2 \leq \|y\|^2 \sum_{n \geq 1} 4^{-n},$$

so $T_0 : E \rightarrow \ell_2$ is bounded. Its restriction to F is injective because all a_n are nonzero and (f_n) is total. Theorem 1 applies.

If F is separable, choose a dense sequence (u_n) in its unit sphere and, by Hahn–Banach, choose $f_n \in F^*$ with $\|f_n\| = f_n(u_n) = 1$. For nonzero $x \in F$, some u_n lies within $1/2$ of $x/\|x\|$, and then

$$f_n(x/\|x\|) > f_n(u_n) - \frac{1}{2} = \frac{1}{2}.$$

Thus (f_n) is total. $\square$

Remark 3. *Theorem 1 also recovers the source’s Hilbert case, including nonclosed F : take $H = E$ and T_0 to be the orthogonal projection onto $\overline{F}$, which restricts to the identity on F .*

5 Scope, upgrade obstruction, and novelty

The theorem covers all separable F and also nonseparable spaces possessing a countable total dual family. For a completely arbitrary nonseparable F , an uncountable total family does not automatically yield a bounded map into $\ell_2(I)$: a summable positive weight family has countable support, while unweighted evaluations need not be square summable. More generally, (b3) already requires a bounded bilinear form that is right-nondegenerate on F . No universal construction or explicit counterexample was found, so the full arbitrary- F question remains open.

A bounded search through 11 August 2026 covered the run indexes; exact arXiv id, title, question, and authors; follow-up and citation queries; non-Hilbert maximum displacement; Hilbert-separating operators; countable total dual families; and nondegenerate bounded bilinear forms. No later answer or this criterion was found.

References

- [1] N. N. Thieu and N. D. Yen, *The Hartman–Stampacchia Theorem and the Maximum Displacements of Nonvanishing Continuous Vector-Valued Functions*, arXiv:2502.00746v1 (2025), especially Question 6 and Theorems 4.5–4.6.